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Diabetic Foot Ulcer using Machine Learning

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***Abstract*- Diabetic foot ulcers (DFUs) are a significant and life-threatening complication of diabetes, affecting 15% to 25% of diabetic individuals worldwide. If not diagnosed and managed early, DFUs can lead to severe infections, prolonged hospitalization, and lower limb amputations. Traditional diagnostic methods on clinical examination and subjective assessment, may result in delayed intervention. This study presents a machine learning-based framework for the automated detection and classification of DFUs to assist in early diagnosis and timely treatment. A dataset of 1,055 images was used, comprising 512 normal (healthy) foot images and 543 abnormal (DFU-affected) images, sourced from Kaggle. The study evaluates four machine learning models Support Vector Machines (SVM), Random Forest (RF), Logistic Regression (LR), and AdaBoost to classify images into normal and DFU categories. Performance evaluation was conducted using key metrics such as accuracy, precision, recall, F1-score, and specificity. The results demonstrate that SVM and LR achieved the highest accuracy of 98%, outperforming RF and AdaBoost. These findings suggest that machine learning can serve as an efficient and reliable diagnostic tool, reducing the dependency on manual assessment and enabling faster, more objective, and cost-effective DFU detection. Future work will focus on improving model generalization using larger datasets and advanced deep-learning techniques.**

Keywords: Diabetic foot ulcers, machine learning, early detection, prognosis, support vector machines (SVM), logistic regression (LR), random forest (RF), AdaBoost, classification accuracy, healthy skin, abnormal skin.

I. INTRODUCTION

Diabetic foot ulcers (DFUs) are a common complication associated with diabetes. These ulcers develop as a result of a combination of factors, including peripheral neuropathy, which reduces sensation in the feet, and peripheral arterial disease, which impairs blood circulation. DFUs have the potential to infect, which may result in the removal of the foot or lower limb [1]. The condition negatively impacts their quality of life, social participation, and livelihood [2]. Diabetic foot affects 15% to 25% of individuals with diabetes, often leading to severe foot ulcers that result in lower limb amputation, hospitalization, and, in the absence of proper treatment, may in severe cases lead to the patient's death [3].

In 2019, the International Diabetes Federation reported that around 463 million adults globally were affected with diabetes. By 2045, this number is predicted to rise to 700 million. People with diabetes have a 34% lifetime risk of developing a diabetic foot ulcer (DFU), which means that 1 in 3 individuals with diabetes will be affected by DFU throughout their lifetime [4]. Each year, approximately 67% of amputations in the US and 90% of amputations in the UK are caused by diabetes [5].

DFUs are often characterized by poor wound healing, which can progress to lower limb amputation and adversely impact survival outcomes. Peripheral vascular and peripheral neuropathy disease are recognized as the most critical risk factors for the development of DFUs in diabetic patients [6].

Computer vision and ML algorithms have recently gained prominence in disease diagnosis and prognosis across various medical imaging techniques, such as CT, MRI, dermatology, WSI, and ultrasound. In DFU assessment, DL approaches are widely used for detecting and identifying DFUs [7]. Furthermore, researchers have combined traditional ML methods to analyze DFU images [8].

II. RELATED STUDY

Alzubaidi et al. [9] developed a model for detecting DFUs more quickly and cost-effectively. The proposed deep learning (DL) model, DFU_QUTNet, automatically classifies images using a dataset of 754-foot images, including both healthy skin and DFU-affected skin. The DFU_QUTNet model achieved an F1 score of 94.5%. This study highlights a more accessible and efficient system for diagnosing DFUs.

Nagaraju et al. [10] conducted a study utilizing deep learning (DL) to automatically detect and classify DFUs from medical images. They proposed the SSODL-DFUDC model for DFU classification and detection. The model was trained on the Kaggle dataset, which comprises 844 samples, including 434 normal and 410 abnormal images. The SSODL-DFUDC method achieved an accuracy, precision, recall, and F1 score of 99.29%. In conclusion, the proposed model demonstrated higher accuracy compared to other techniques, making DFU detection faster and more efficient.

Similarly, Goyal et al. [11] proposed a machine learning (ML) approach to analyze the differences and developed a new deep learning (DL) model for DFU detection using a large database of foot images. In this study, DFUNet utilizes convolutional neural networks (CNN) to extract features. The DFUNet achieved an AUC score of 96%, outperforming other methods. This approach could

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make DFU detection faster, more cost-effective, and more accessible, thereby improving care for diabetic patients.

Garaawi et al. [12] recently focused on automated methods using computer vision and machine learning (ML) to improve DFU detection. The dataset used was collected from recently published DFU datasets. While most current methods employ convolutional neural networks (CNN) with RGB images, this study enhances accuracy by combining texture features with RGB images. These methods achieved an AUC of 98.1% and F-measures of 95.2%, making them more effective at identifying DFUs.

Liu et al. [13] developed a deep learning (DL) model to detect infection and ischemia in DFU using images from the DFU dataset. The proposed model, EfficientNet, was used to classify infection and ischemia, and was compared to other models. EfficientNet achieved the highest accuracy, with 99% for ischemia and 98% for infection. EfficientNet is a more accurate system for detecting DFU infection and ischemia.

Ashan et al. [14] proposed several deep learning (DL) models, including AlexNet, VGG16/19, GoogleNet, ResNet/101, MobileNet, SqueezeNet, and DenseNet, to detect infection and ischemia. The system utilizes the DFU2020 dataset. The ResNet50 model achieved the highest accuracy, with 99.49% for ischemia and 84.76% for infection.

Almufadi et al. [15] focused on classifying DFUs into two types: ischemia and infection. The study used deep learning (DL) models such as EfficientNetB0, DenseNet121, and ResNet101 to classify DFUs. Machine learning (ML) models, including logistic regression and AdaBoost, were applied to improve the accuracy of the models. EfficientNetB0 with logistic regression achieved an accuracy of 97% for ischemia, while AdaBoost achieved 93% accuracy for infection.

Khandakar et al. [16] focused on machine learning (ML) and convolutional neural network (CNN) models to detect diabetic feet. The proposed method uses MobileNetV2 and the AdaBoost classifier. The models achieved an F1 score of 95% for MobileNetV2 in classifying thermogram images, and 97% for the AdaBoost classifier using 10 key features. The system is fast and can be used on mobile apps, allowing individuals to assess DFU risks at home.

Pushpa Leela et al. [17] discussed applying classification techniques such as SVM, Naive Bayes, KNN, C4.5, and decision trees to predict DFU in type-2 diabetic patients. The dataset applied was a real-world collection from a hospital, comprising data from 455 individuals, of whom 355 were diabetic patients and 100 were non-diabetic. After applying all the techniques, the results were compared based on accuracy, with SVM outperforming the other algorithms, achieving an accuracy of 92.22%.

Ahmed [18] has proposed a model for diabetes prediction based on medical history. The model uses real-world data from the Health Facts database to predict whether a patient has diabetes. The model checks whether the HbA1C value is under control or not. Techniques used for data mining include Naive Bayes, logistic regression, and J48 for the predictive model. After comparing the models, logistic regression (LR) showed the best performance, with accuracy, F-measure, precision, and recall values of 74.4%, 65.3%, 73%, and 7.44%, respectively.

III. MATERIALS AND METHODS

A. Dataset selection and distribution

The study utilizes a dataset of healthy foot images and diabetic foot ulcer images available in the Kaggle repository. The dataset includes normal and abnormal images, with a total of 1055 images. Specifically, 512 normal images and 543 abnormal image categories were utilized. The foot images are derived from a comprehensive database maintained by the DFU images.

Table 1. Dataset splitting

Classes	Images
Normal	512
Abnormal	543
Total	1055

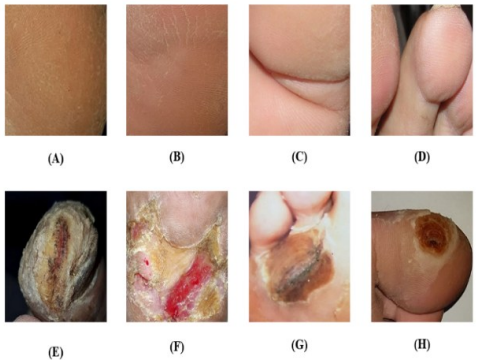

Fig 1. Diabetic foot images (A) – (D) normal foot. (E) – (H) abnormal foot.

B. Image embedding

Image embedding is the process of converting an input image into a compact, high-dimensional vector representation using a deep learning model. In the interface provided, the selected image attribute, labeled as "S image," is processed using Inception v3, a convolutional neural network pre-trained on the ImageNet dataset. This model extracts meaningful features like patterns, textures, and shapes, which are essential for tasks such as classification and similarity detection. The resolution of the input image is set to 1055 x 1055 pixels. The model is applied to it to ensure compatibility. The interface also has an Apply Automatically that makes the embedding automatically go through once the input or configuration change is affected on this interface. This works towards providing the image's concise, informative description and helps in improving accuracy with machine learning tasks.

C. Proposed framework

SVM Model

Support Vector Machines (SVMs) are supervised learning models used for classification and regression tasks. SVMs map input features into a higher-dimensional space to find the optimal hyperplane that separates classes. The margin between classes is maximized, enhancing generalization. SVMs are particularly effective when the relationship between features is non-linear, leveraging kernel

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functions like radial basis functions (RBF) and polynomial kernels to achieve.

The proposed SVM model employs a radial basis function (RBF) kernel to handle non-linearity. It optimizes the hyperplane using a regularization parameter *CC*C set to 1.0 and a kernel coefficient to balance the trade-off between accuracy and generalization. This configuration enables robust classification for datasets with overlapping features. Loss minimization is achieved through hinge loss, ensuring margin maximization while reducing misclassification.

Random Forest Model

Random Forests are ensemble learning methods that construct a multitude of decision trees during training. Each tree in the forest is built using a random subset of features and samples, introducing diversity to reduce overfitting. Predictions are made by aggregating the outputs of all trees through majority voting for classification or averaging for regression tasks.

The proposed Random Forest model consists of 100 decision trees, each trained on bootstrapped samples of the dataset. The maximum depth of each tree is limited to 10 to control complexity and avoid overfitting. Splitting criteria use the Gini impurity metric, ensuring optimal feature selection at each node. Feature importance analysis is performed to highlight the most influential variables. This model strikes a balance between computational efficiency and predictive performance.

AdaBoost Model

Adaptive Boosting (AdaBoost) is an ensemble technique that combines multiple weak classifiers to create a strong classifier. Each weak classifier focuses on the samples misclassified by previous classifiers, improving the model's ability to handle difficult cases. The final prediction is a weighted sum of all classifiers' outputs.

The proposed AdaBoost model uses decision stumps as weak classifiers, with 50 iterations to build the ensemble. A learning rate of 0.1 controls the contribution of each weak classifier, preventing overfitting. The model adjusts sample weights dynamically to emphasize misclassified instances, enhancing overall accuracy. Performance is evaluated using the exponential loss function, which aligns with AdaBoost's iterative optimization process.

Logistic Regression Model

Logistic Regression is a linear model designed for binary and multi-class classification. It estimates the probability of class membership by modeling the log odds as a linear combination of input features. The sigmoid function transforms these log odds into probabilities, making the model suitable for tasks with dichotomous outcomes.

The proposed Logistic Regression model employs a regularization term to prevent overfitting, with a penalty parameter *CC*C set to 1.0. Features are normalized to a mean of zero and unit variance for efficient convergence during optimization. The model is trained using the stochastic gradient descent (SGD) optimizer, with a learning rate Cross-entropy loss is employed to evaluate performance, ensuring accurate probability estimations for class assignments.

D. Model evaluation metrics

To assess the performance of the proposed method five commonly used metrics in classification applications are evaluated accuracy, F1-score, precision, recall, and specificity.

Accuracy is a fundamental metric that measures the proportion of correctly classified instances out of the total predictions, providing an overall assessment of the model's correctness.

F1 Score can be determined that the F1 score represents a balance that comes with the trade-off from false positives to false negatives by being the harmonic mean between precision and recall.

Precision measures the proportion of correctly predicted positive instances out of all the cases predicted as positive, reflecting the model's ability to avoid false positives.

Recall measures the proportion of actual positive instances that the model correctly classified.

Specificity measures the proportion of correctly identified negative cases. It is the measure of how well the model avoids false positives.

IV. RESULTS AND DISCUSSION

For classifying the DFU, four machine learning models were used: svm, random forest, adaboost and logistic regression. The images were classified into two categories, normal (healthy foot) and abnormal foot ulcer.

Table 2. Performance metrics of AdaBoost, LR, SVM, RF

MODEL	ACC	F1	PREC	RECALL	SPEC
SVM	98.8%	98.8%	98.8%	98.8%	98.8%
RF	97.5%	97.5%	97.5%	9.75%	9.75%
ADA BOOST	97.3%	97.2%	97.3%	97.3%	9.72%
LR	98.7%	98.7%	98.7%	98.7%	98.6%

The table compares the performance of four machine learning classifiers SVM (Support Vector Machine), Random Forest (RF), AdaBoost, and Logistic Regression (LR) across five metrics: Classification Accuracy (CA), F1 Score, Precision (PREC), Recall (RECALL), and Specificity (SPEC). SVM outperforms all other models, achieving 98.8% across all metrics, followed closely by LR with slightly lower values (98.7%). In contrast, Random Forest and AdaBoost demonstrate lower performance, with an accuracy of around 97.5% and significant drops in recall and specificity (9.75% and 9.72%, respectively). This indicates that while RF and AdaBoost perform adequately for accuracy and precision, they struggle with correctly identifying true positives and negatives, likely due to class imbalance or inadequate handling of complex cases.

A. Performance of SVM:

The precision, recall, F1 score, accuracy, and specificity all at 98.8%, the SVM approach has been highly effective, scoring well on all parameters. The model missed only 8 cases of abnormal skin (ulcers) while correctly identifying 507 cases. Only five false positives, correctly identified 535 cases of normal skin, showing high specificity. SVM thus

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proves its robustness in terms of sensitivity and specificity, thus indicating a reliable and successful approach to classifying foot ulcers. Its performance could be further optimized to reach higher diagnostic accuracy levels.

Actual \ Predicted	Abnormal(Ulcer)	Normal(Healthy skin)	Σ
Abnormal(Ulcer)	507	5	512
Normal(Healthy skin)	8	535	543
Σ	515	540	1055

Fig.2.Confusion Matrix of SVM

B. Performance of Random Forest:

At about 97.5%, the Random Forest approach has shown excellent performance by attaining high metrics in terms of accuracy, precision, recall, specificity, and F1 scores. Despite wrongly classifying 14 cases as healthy skin, it correctly classified 498 cases of ulcers. Moreover, it had a very good ability to classify correctly 531 cases of healthy skin, misclassifying only 12 cases as ulcers.

Actual \ Predicted	Abnormal(Ulcer)	Normal(Healthy skin)	Σ
Abnormal(Ulcer)	498	14	512
Normal(Healthy skin)	12	531	543
Σ	510	545	1055

Fig.3.Confusion Matrix of Random Forest

C. Performance of Adaboost:

The AdaBoost technique has achieved precision, recall, and F1 scores of almost 97% each, which is a great performance. It showed good memory in reducing false negatives by correctly detecting 489 ulcer cases while missing 23 cases. The program correctly detected 537 cases of normal foot and misclassified only 6 cases.

Actual \ Predicted	Abnormal(Ulcer)	Normal(Healthy skin)	Σ
Abnormal(Ulcer)	489	23	512
Normal(Healthy skin)	6	537	543
Σ	495	560	1055

Fig.4. Confusion Matrix of Adaboost

D. Performance of Logistic Regression:

The excellent precision, recall, and an F1 score of about 98.7% each, the Logistic Regression approach has been performing very well so far. It showed excellent sensitivity by minimizing false negatives by correctly identifying 503 abnormal (ulcer) cases and missing only 9 cases. With a high specificity of 98.6%, 538 normal instances are correctly identified, with a significant reduction in false positives. Logistic regression is a careful yet effective method for distinguishing between skin conditions.

Actual \ Predicted	Abnormal(Ulcer)	Normal(Healthy skin)	Σ
Abnormal(Ulcer)	503	9	512
Normal(Healthy skin)	5	538	543
Σ	508	547	1055

Fig.5.Confusion Matrix of Logistic Regression

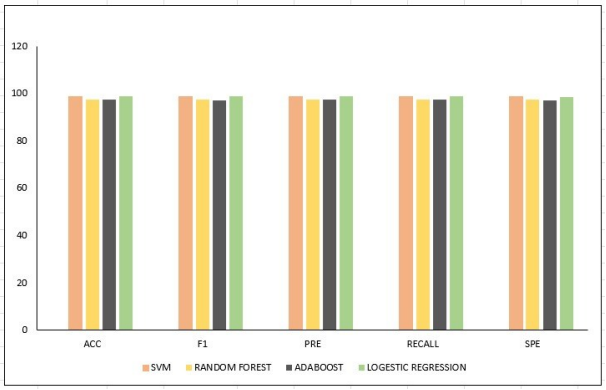

Fig.6. Comparison of classifier performance across metrics

The graph visually compares the performance of four classifiers SVM, Random Forest (RF), AdaBoost, and Logistic Regression (LR) across five metrics: accuracy, F1 score, precision, recall, and specificity. SVM consistently outperforms the other models, achieving the highest values across all metrics. This is followed closely by logistic regression, which shows comparable performance with only minor differences. In contrast, Random Forest and AdaBoost display significantly lower recall and specificity, highlighting their struggles with accurately identifying true positives and true negatives. The graph emphasizes the strong overall performance of SVM, while also indicating that RF and AdaBoost require optimization to handle imbalanced or complex data effectively.

V. CONCLUSION

Diabetic foot ulcers (DFUs) pose severe health risks, often leading to amputations if not detected early. This study evaluated four machine learning models SVM, RF, LR, and AdaBoost for DFU classification, with SVM and LR achieving the highest accuracy (98%). Despite promising results, limitations such as dataset size and the need for more advanced deep-learning techniques remain. Real-world challenges include computational efficiency and clinical adoption, requiring models that are both lightweight and interpretable. Future work should focus on expanding datasets, improving real-time deployment, and integrating multi-modal diagnostic approaches. With continued advancements, AI-based DFU detection can significantly improve early diagnosis and patient outcomes in healthcare.

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